

HATS-2b

R-2207

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATS-2_250415 · created 2026-08-14T01:16:23+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460780.7138 +/- 0.0045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.176 +/- 0.051
Transit depth (Rp/Rs)^2	3.09 +/- 1.8 [%]
Orbital Inclination (inc)	83.34 +/- 2.06
Airmass coefficient 1 (a1)	0.7 +/- 0.3
Airmass coefficient 2 (a2)	0.13 +/- 0.33
Scatter in the residuals of the lightcurve fit is	36.95 %
Best Comparison Star	#3 - [512, 425]
Optimal Aperture	1.82
Optimal Annulus	6.29
Transit Duration (day)	0.0763 +/- 0.0099

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.176 ± 0.051 vs 0.1335 (deviation 0.67σ)
Duration fit vs published	0.0763 d vs 0.0871 d (deviation 0.88σ)
Mid-time O–C	-7.1 min
Depth SNR	2.8
Residual scatter	36.95 %

Light curve

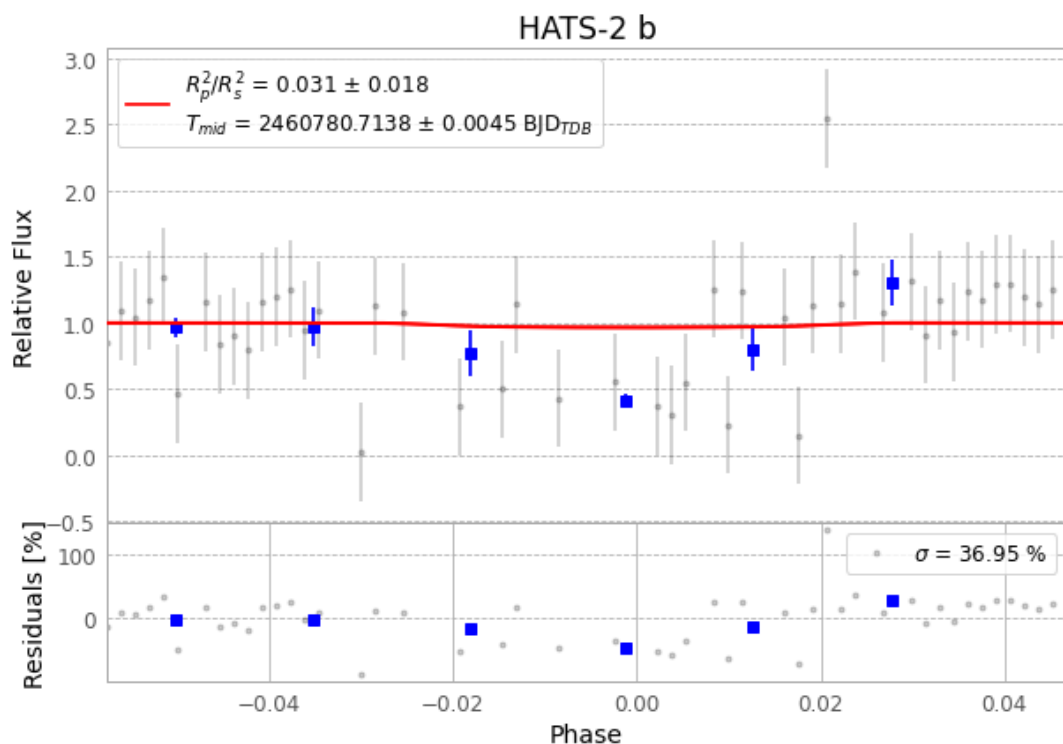


Figure 1 — FinalLightCurve_HATS-2 b_2025-04-14.png (SHA-256 a946e9d69262552...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [332, 202], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 af925bd93cda9104...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

69 FITS frames (45 MB), HATS-2250415030616.FITS → HATS-2250415063015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATS-2-250415.log (78328236e251...), FinalLightCurve_HATS-2 b_2025-04-14.png (a946e9d69262...), FinalLightCurve_HATS-2 b_2025-04-14.csv (16a3283927da...)

Compute resources

Wall time 169.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-14 01:16Z.